

From Energy-Based Memory to Deep Representation: RBM and DBN as the Bridge to Deep Learning

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Introduction: Three Guiding Questions

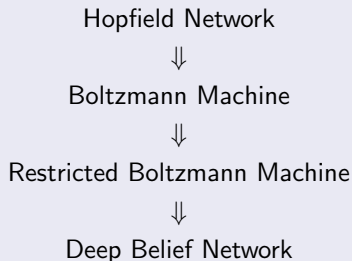
- ① Why did the field move from Hopfield/Boltzmann models to RBMs?
- ② What does an RBM actually learn?
- ③ Why were DBNs important in the history of deep learning?

Today's focus

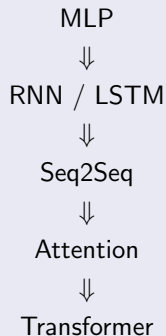
We trace how neural computation moved from **energy-based memory** to **trainable deep representations**.

Two Lineages: Revival vs. Architecture

1. Conceptual lineage of deep learning revival



2. Architectural lineage toward Transformers



Key distinction

DBNs helped show that **deep models could be trained** through greedy layer-wise pretraining, although they are not the direct architectural ancestor of Transformers.

Recap: What Did Hopfield and Boltzmann Give Us?

Hopfield Network

- Recurrent network for **associative memory**
- Stable patterns correspond to **energy minima**
- Retrieval as movement toward an attractor

Boltzmann Machine

- Stochastic energy-based model
- Introduces **hidden units**
- Learns a probability distribution over visible data

Shared idea

Cognition can be modeled as movement in an **energy landscape**: memory, inference, and generation become probabilistic dynamics.

The Problem Before RBM/DBN

- ❶ **Boltzmann Machines were theoretically elegant, but hard to train.**
 - Fully connected stochastic networks require expensive sampling.
 - Hidden units interact with each other, making inference difficult.
- ❷ **Deep neural networks were widely viewed as difficult to optimize.**
 - Deep supervised backpropagation often got stuck or learned poorly.
 - Learning useful multi-stage feature extractors was often considered infeasible.

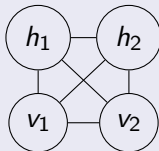
Historical context

In the late 1990s, neural nets and backpropagation were largely forsaken by the machine-learning community and ignored by the computer-vision and speech-recognition communities. It was widely thought that learning useful, multistage, feature extractors with little prior knowledge was infeasible. In particular, it was commonly thought that simple gradient descent would get trapped in poor local minima — weight configurations for which no small change would reduce the average error. In practice, poor local minima are rarely a problem with large net works.

From Boltzmann Machine to RBM

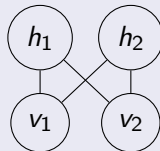
Boltzmann Machine

- Visible-visible connections: **possible**
- Hidden-hidden connections: **possible**
- Visible-hidden connections: **possible**



Restricted Boltzmann Machine

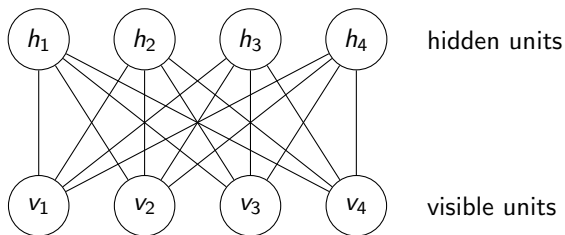
- Visible-visible connections: **no**
- Hidden-hidden connections: **no**
- Visible-hidden connections: **yes**



The restriction is the trick

By removing intra-layer connections, the model becomes much easier to sample from and train.

RBM Architecture



- **Visible units:** observed data, such as pixels or binary features
- **Hidden units:** latent feature detectors
- **No intra-layer connections:** no $v-v$ and no $h-h$ edges
- **Bipartite undirected graph:** symmetric connections between visible and hidden layers

Why the Restriction Matters: Conditional Independence

Given visible vector $\mathbf{v}$

Hidden units become conditionally independent:

$$p(\mathbf{h} \mid \mathbf{v}) = \prod_j p(h_j \mid \mathbf{v})$$

Given hidden vector $\mathbf{h}$

Visible units become conditionally independent:

$$p(\mathbf{v} \mid \mathbf{h}) = \prod_i p(v_i \mid \mathbf{h})$$

Practical consequence

To generate data from an RBM, we can start with a random state in one of the layers and then perform alternating Gibbs sampling: All of the units in one layer are updated in parallel given the current states of the units in the other layer and this is repeated until the system is sampling from its equilibrium distribution.

Reference: Hinton, Osindero, and Teh, *Neural Computation*, 2006.

Energy Intuition

RBM energy function

$$E(\mathbf{v}, \mathbf{h}) = -\mathbf{a}^\top \mathbf{v} - \mathbf{b}^\top \mathbf{h} - \mathbf{v}^\top \mathbf{W} \mathbf{h}$$

- $\mathbf{a}$: bias of visible units
- $\mathbf{b}$: bias of hidden units
- $\mathbf{W}$: visible–hidden connection weights
- Lower energy means a more compatible pair of visible data and hidden features.

Intuition

The RBM learns an energy landscape where training data have low energy, while unlikely configurations have higher energy.

Learning RBM: Positive Phase vs. Negative Phase

Positive phase

- Clamp the visible units to real data.
- Infer hidden activations.
- Increase visible–hidden correlations that explain the data.

$$\langle v_i h_j \rangle_{\text{data}}$$

Negative phase

- Let the model run by Gibbs sampling.
- Observe what the model generates at equilibrium.
- Decrease correlations produced by the model itself.

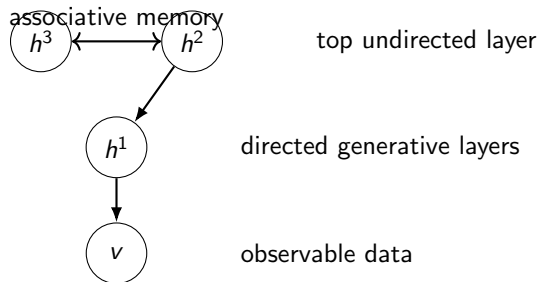
$$\langle v_i h_j \rangle_{\text{model}}$$

Maximum-likelihood learning signal

$$\Delta w_{ij} \propto \langle v_i h_j \rangle_{\text{data}} - \langle v_i h_j \rangle_{\text{model}}$$

Reference: Hinton, Osindero, and Teh, 2006.

What Is a Deep Belief Network?



Definition from the DBN perspective

A DBN has an **undirected associative memory** at the top and a **directed generative model** below it, transforming high-level representations into observable variables such as image pixels.

Reference: Hinton, Osindero, and Teh, 2006.

DBN Architecture in the Paper

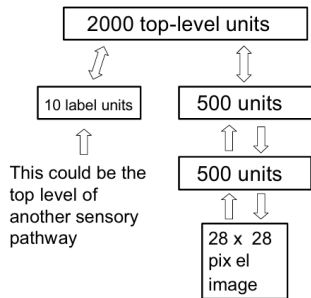


Figure: Deep Belief Network architecture

- Top two hidden layers form an undirected associative memory.
- Lower layers form a directed generative model.

Why Deep Networks Were Hard

Core difficulty: posterior inference

In a deep directed belief network, learning requires inferring which hidden states should explain the observed data.

- Many hidden causes can explain the same observation.
- This creates **explaining away**: once one cause explains the data, alternative causes become less likely.
- The posterior over hidden activities becomes hard to compute.

Example

Observation: *The house jumped.*

- Cause 1: earthquake
- Cause 2: truck hit the house

If earthquake explains it, the truck hypothesis becomes less likely.

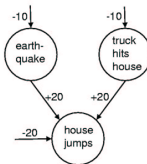


Figure: A simple logistic belief net

Complementary Priors: Mention, Do Not Prove

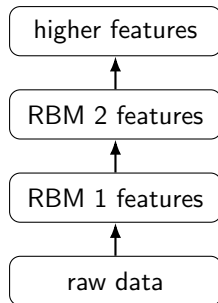
Main idea

A **complementary prior** cancels the explaining-away correlations created by the likelihood, making the posterior approximately factorial.

- Explaining away makes hidden causes statistically dependent.
- The complementary prior is designed to offset that dependence.
- This creates a bridge between RBMs and deep directed models.
- It provides theoretical support for greedy layer-wise learning.

Greedy Layer-Wise Pretraining

- 1 Train an RBM on raw data.
- 2 Freeze the first layer.
- 3 Use hidden activations as “data” for the next RBM.
- 4 Repeat layer by layer.
- 5 Fine-tune the full model.



Key mechanism

A complicated deep model is learned as a sequence of simple models. Each layer creates a nonlinear representation used as input for the next layer.

Reference: Hinton, Osindero, and Teh, 2006, Section 4.

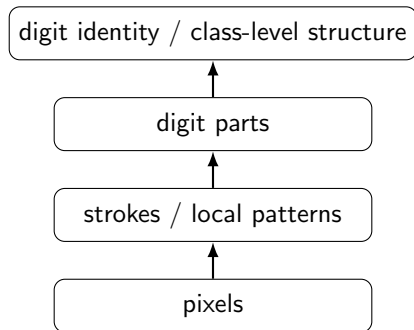
Pretraining vs. Fine-tuning

Stage	Learning method	Purpose	Intuitive analogy
1. Pretraining	RBM / DBN <i>Unsupervised</i>	Initialize weights to meaningful values	Draw the rough sketch <i>feature extraction</i>
2. Fine-tuning	Backpropagation <i>Supervised</i>	Adjust the model for a specific task, such as classification	Add colors and finish the artwork for the target purpose

Key idea

Pretraining gives the network a good starting representation; fine-tuning adapts that representation to the final task.

DBN as Hierarchical Representation Learning



Why this mattered

DBNs made deep learning feel plausible again: each layer could learn a more abstract representation without requiring labels at every step.

Looking Into the Mind of the Network

Generative interpretation

A DBN is not only a discriminative system that predicts labels. It can also **generate** data from its internal representations.

- Run Gibbs sampling in the top-level associative memory.
- Send the sampled representation downward through generative connections.
- Generate visible images from hidden states.

Reference: Hinton, Osindero, and Teh, 2006, Section 7.

Interpretability by generation

To understand what a hidden representation means, ask:

“What image does it generate?”

Deep Learning Revival: What Changed Around 2006?

- Deep feedforward networks regained attention around 2006 through **unsupervised learning procedures**.
- These procedures learned feature detector layers without requiring labeled data.
- Pretraining initialized deep-network weights to sensible values.
- Standard backpropagation could then fine-tune the whole network.

Historical significance

RBMs and DBNs were not the final form of modern deep learning, but they helped change the field's belief about whether deep models could be trained at all.

Reference: LeCun, Bengio, and Hinton, *Nature*, 2015.

Takeaway

- ① Hopfield and Boltzmann models framed cognition as **energy-based memory and probabilistic dynamics**.
- ② RBMs made Boltzmann-style learning more practical through **restricted connectivity and conditional independence**.
- ③ DBNs showed that deep models could be trained through **greedy layer-wise pretraining**, helping revive deep learning.

Bridge to the larger study theme

From brain-inspired memory models to deep representation learning, we see one recurring question:

How can a system learn useful internal representations of the world?

- ① Should interpretability be part of intelligence?
- ② Are brain-inspired models still necessary for AI progress?
- ③ What should count as a “good representation”?

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